

ALHAZEN

An innovator in the field of optics, Alhazen investigated how light is reflected and refracted, and was one of the first experimental scientists.

Born in the Persian city of Basra (in modern-day Iraq) in c.965, Alhazen was educated in Baghdad during the Golden Age of Islamic civilization. He worked as a civil servant and pioneered the now-standard scientific method of proposing hypotheses and then testing them through experiments.

Under house arrest in Egypt, where he had promised to dam the River Nile for the Caliph but failed to do so, Alhazen began his seven-volume *Book of Optics*. He disputed the theories of Euclid and Ptolemy that “rays” are emitted from the eye and bounce back from an object. Instead, he experimented with bulls’ eyes and found that light enters a hole (the pupil) and is focused by a lens onto a sensitive surface (the retina) inside the eye. Alhazen also studied the properties of glass, mirrors, and lenses and showed that light travels in straight lines.



“The **seeker after truth** ... submits to **argument and demonstration.**”

Alhazen, c.1025

MILESTONES

EARLY TEXTS

Writes his first papers in c.1000 on subjects including astronomy, optics, and mathematics.

DOOMED PROJECT

Tasked with regulating the flow of the River Nile in c.1011, but it proves impossible.

FOCUSED STUDY

Resigns from his civil-service job in 1027 to devote himself full-time to scientific study.